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COURSE 1.1 / COMPANION WORKBOOK

Windows Server 2025

DC01 and DC02

Build the identity foundation

A practical companion for configuring Active Directory and DNS, checking the result, and recording evidence.

MATERIALS EDITION 0.4 / COURSE 1.1

Updated 8 October 2026 · sbd-dc01-dc02

NAME _____ / DATE _____

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Use it alongside the course

The interactive companion adds chapter progress, knowledge checks, evidence notes, and a 3D view of the network as it takes shape. The workbook is your readable reference and writing space.

YOUR PACE Read the foundations first if these concepts are new. Return to them whenever a configuration choice needs a clearer explanation.

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What is Active Directory?

The shared directory behind domain identities.

A directory for people and computers

Active Directory Domain Services (AD DS) is the Windows Server directory service. It stores objects such as user accounts, computer accounts, and groups, together with their attributes. Administrators use this shared directory to organize identities and manage access consistently.

The domain and the controller

A domain is a logical collection of directory objects, named here sbd.test. A domain controller (DC) is a server running AD DS that stores and serves that directory. DC01 and DC02 are two servers serving the same domain. A forest contains one or more domains; this build starts with one.

Identity, permissions, and policy

Authentication checks who is signing in. Authorization determines what that identity may access. Signing in successfully does not grant access to every resource: the destination service still applies its permissions. Groups make it practical to assign access to roles instead of managing every person separately.

Organization and consistent settings

Organizational units (OUs) organize objects and support delegated administration and Group Policy. Group Policy applies centrally managed settings to users and computers. A global catalog helps find directory objects across a forest; both controllers in this course provide that role.

REMEMBER AD DS provides directory and identity services. DNS helps clients find those services. Each has a distinct job.

THINK IT THROUGH Explain the difference between the sbd.test domain and the SBD-DC01 server.

Microsoft · What is Active Directory Domain Services?
Microsoft · Group Policy and organizational units
Microsoft · Authentication and authorization

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What is DNS?

The name and service directory that makes AD discoverable.

Names become reachable addresses

Domain Name System (DNS) answers queries about names. An A record maps a hostname to an IPv4 address; an AAAA record maps it to an IPv6 address. In this sample, SBD-DC01.sbd.test resolves to 10.0.10.10. DNS also works on an isolated network without internet access.

AD needs service discovery

Service (SRV) records advertise which servers provide a service. A client uses DNS to discover suitable domain controllers, resolves their hostnames, and contacts a controller for authentication. DNS does not validate the password. A running DC can still be difficult to find if the DNS records or client settings are wrong.

What a zone contains

A DNS zone holds records for a portion of the namespace. With an AD-integrated zone, records are stored in AD DS and replicated to DNS servers within the configured replication scope. Secure dynamic updates restrict which identities may change records. Check the zone scope and update policy on the completed build.

Two roles, two kinds of checks

Installing DNS and installing AD DS provide different services, even when they run on the same server. Query each DNS server for SRV and host records, then check directory replication separately. A correct DNS answer proves neither a successful sign-in nor access to a protected resource.

REMEMBER Discovery sequence: ask DNS for a domain service → resolve the server name → contact a reachable DC → authenticate.

THINK IT THROUGH If you can reach 10.0.10.10 but cannot discover sbd.test, which DNS records and client settings would you inspect?

SEE IT INSIDE THE NETWORK

Sign in, step 2: the workstation asks DNS for the same domain controller locator records you query in Verify.

securebydesigninc.com/inside-the-network#process=signin&step=2

Microsoft · What is DNS?
Microsoft · Active Directory-integrated DNS zones
Microsoft · Locating domain controllers

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Why use two domain controllers?

Keep identity and DNS services available when one server is unavailable.

Remove the single-server dependency

With one DC, its outage removes the only live directory service point. With two healthy, reachable controllers, clients can discover a surviving DC for supported operations. This also lets you patch or restart one server while the other continues serving requests. Reconnection may take time; uninterrupted service is not guaranteed.

Both controllers do useful work

Both writable DCs can process normal directory changes and authentication requests. Replication shares directory changes between them. They are not a primary server and an idle spare. Certain operations still depend on a single FSMO role holder, so role placement and recovery need their own plan.

Make the redundancy real

Microsoft recommends at least two virtualized DCs per domain on different virtualization hosts. Two VMs on one host share that host as a failure point. Consider shared storage, power, and network dependencies too. Run DNS on both here, verify zone replication, and configure clients to use both internal DNS addresses.

Availability still needs recovery

Replication can spread an accidental deletion or harmful change to both servers. Keep protected backups and test recovery separately. Two is a starting point for this design, not a universal capacity target: additional sites, users, and recovery needs may require more controllers.

REMEMBER Before relying on DC02, verify replication and DNS, review shared dependencies, and test client discovery during an approved maintenance window.

THINK IT THROUGH If both controllers run on the same physical host, which outage would still stop both services?

SEE IT INSIDE THE NETWORK

What if DC01 is down? The query to DC01 gets no answer, the workstation retries with SBD-DC02, and sign-in continues.

securebydesigninc.com/inside-the-network#process=dc-down

Microsoft · Avoiding single points of failure
Microsoft · Operations master role placement
Microsoft · Locating domain controllers
Microsoft · DNS client configuration

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What crosses the network?

The ports a domain member needs to reach its domain controllers.

Your lab is one flat network, so nothing stands between a client and DC01. In a segmented network, every one of these conversations has to be allowed on purpose. Inside the Network shows them crossing the core switch.

PURPOSE	PORT	WHAT IT CARRIES
Name and service discovery	TCP/UDP 53	DNS, including the SRV records that locate domain controllers
Authentication	TCP/UDP 88	Kerberos tickets
Password change	TCP/UDP 464	Kerberos password changes and resets
Directory	TCP/UDP 389 · TCP 636	LDAP and LDAPS; UDP 389 carries the domain controller locator ping
Global catalog	TCP 3268 · 3269	Forest-wide searches; 3269 uses TLS
Group Policy and SYSVOL	TCP 445	SMB reads of policies and scripts from domain controller shares
RPC services	TCP 135 · 49152–65535	Endpoint mapper, then the dynamic range for NetLogon, LSA, and SAM
Time	UDP 123	W32Time; members follow the domain time hierarchy
Administration	TCP 9389	AD Web Services for PowerShell and Administrative Center, from administration hosts only

REMEMBER Allowing DNS and Kerberos isn't enough. Finding a domain controller, applying Group Policy, and reading SYSVOL also need LDAP, SMB, and RPC. In the Inside the Network design, rule F03 permits this directory profile for domain members, with LDAP signing and channel binding required.

THINK IT THROUGH A workstation can resolve names and get Kerberos tickets, but Group Policy never applies. Which ports would you check first?

SEE IT INSIDE THE NETWORK

Sign in, step 4: after it authenticates, the workstation reads directory information and Group Policy from SBD-DC01.

securebydesigninc.com/inside-the-network#process=signin&step=4

Microsoft · Service overview and network port requirements for Windows

Microsoft · Configure a firewall for Active Directory domains and trusts

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01 / PREPARE

Plan your build

Build two new domain controllers in one isolated network. The interactive guide contains the detailed steps, verification commands, and chapter evidence fields.

What you will learn

- Configure DC01 and add DC02 to the same domain.
- Inspect DNS records, service state, and replication health.
- Connect configuration choices to security reasoning and evidence.

Before you begin

Use two new Windows Server 2025 Desktop Experience VMs, console access, and approved installation media. Record your starting OS builds. Keep this exercise isolated; sample domain values must match your own environment plan.

Sample configuration

SETTING	SAMPLE VALUE
Domain and NetBIOS	sbd.test / SBD
First controller	SBD-DC01 / 10.0.10.10/24
Second controller	SBD-DC02 / 10.0.10.11/24
DNS before promotion	DC01: own .10 address / DC02: DC01
Network and time	Isolated subnet / record the internal time source

CM-2 BASELINE Record the intended values and the observed starting configuration. Keep the baseline aligned with changes to the build.

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Build DC01

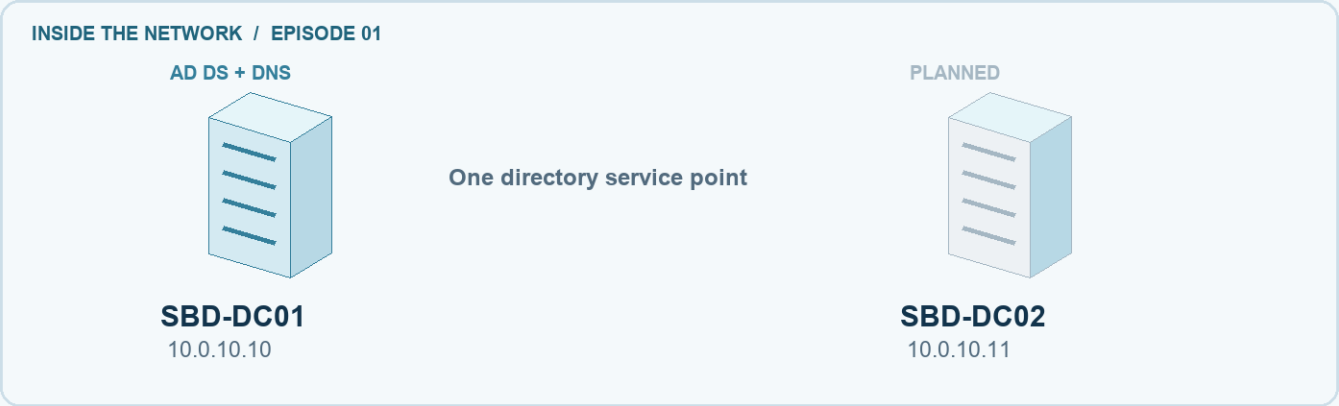
WHY THIS MATTERS Create the first directory and DNS service for the sample forest.

Follow the build

1. On DC01, add the Active Directory Domain Services role and its management tools through Server Manager.

2. Promote the server using Add a new forest and sbd.test. Select DNS, global catalog, Windows Server 2025 functional levels, and a unique DSRM password.

3. Review DNS options, the SBD name, paths, and prerequisites. Resolve blocking errors, install, and restart.



NETWORK STATE DC01 configured; DC02 planned.

Check your result

```
Get-ADDomain | Select-Object DNSRoot, NetBIOSName
Get-Service NTDS,DNS,Netlogon
Get-SmbShare -Name SYSVOL,NETLOGON
```

Expect the planned domain and running services, with both shares present. Investigate missing results before adding DC02.

CM-6 SETTINGS Record the promotion choices and why they fit the environment. Keep passwords out of evidence.

COMMON MISTAKE Role installation alone does not complete domain-controller promotion.

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03 / EXTEND

Add DC02

WHY THIS MATTERS Add another writable controller to the same domain.

Follow the build

1. Point DC02 at DC01 for DNS. Check domain discovery, the DC01 hostname, network reachability, and clock alignment.

2. Install AD DS, then choose Add a domain controller to an existing domain. Use sbd.test and authorized domain credentials. Include DNS and global catalog.

3. Review site, replication source, DSRM password, and prerequisites. Install and restart. Keep DC01 as DC02 DNS until initial replication is healthy.



NETWORK STATE Two controllers with a replication relationship.

Check your result

```
Get-ADDomainController -Filter *
repadmin /replsummary
repadmin /showrepl SBD-DC02
```

Expect both controllers and successful recent replication. Record unresolved errors and the final internal DNS resolver order.

CP-9 RECOVERY A second controller supports availability. Backups and tested recovery remain separate work.

COMMON MISTAKE Choose the existing-domain option. A new forest would create a separate directory.

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Check DNS and replication

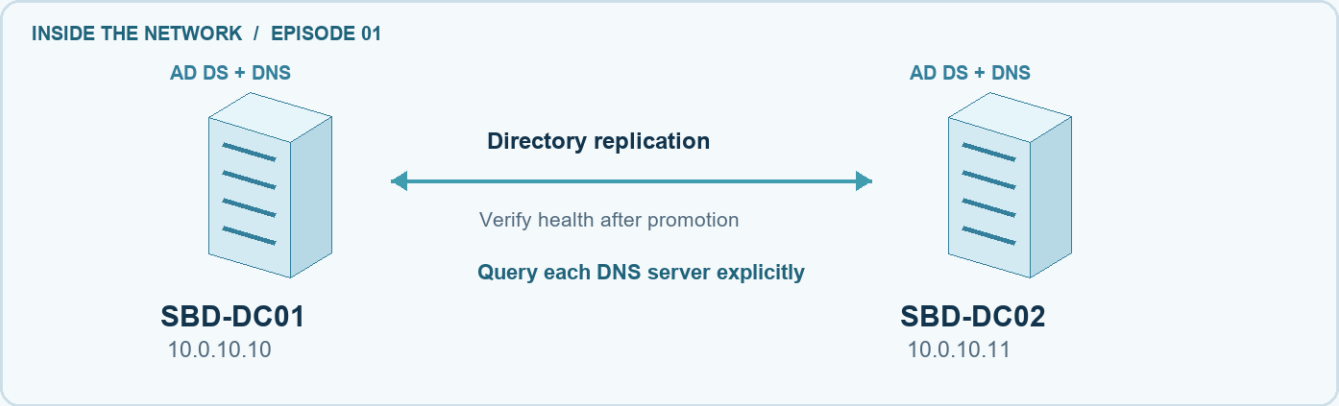
WHY THIS MATTERS Ask each service for an observable result.

Follow the build

1. On a server with the management tools, query the domain locator SRV records against 10.0.10.10 and 10.0.10.11 separately.

2. Resolve both controller hostnames against each resolver. Expect DC01 at 10.0.10.10 and DC02 at 10.0.10.11.

3. Review DCDiag DNS tests and replication results. Capture the queried server, collection time, actual output, and any failures.



NETWORK STATE **Query both DNS services and compare results.**

Check your result

```
Resolve-DnsName '_ldap._tcp.dc._msdcs.sbd.test' `
  -Type SRV -Server 10.0.10.10
dcdiag /test:DNS /e /v
repadmin /replsummary
```

Repeat the DNS query using 10.0.10.11. Use Verification-Commands.txt for the complete query loop and host-record checks.

IA-2 DEPENDENCIES DNS supports identity discovery. These checks do not establish all authentication requirements.

COMMON MISTAKE A successful ping does not demonstrate DNS, replication, or authentication health.

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Optional: see it on the wire

Packet Monitor (pktmon) is built into Windows Server 2025. Run it on SBD-DC02 in an elevated PowerShell window while you query DC01, so the request crosses the network. Then compare the capture with the packet card for Sign in, step 2 in Inside the Network.

```
New-Item -ItemType Directory C:\Evidence -Force
pktmon filter remove
pktmon filter add DNS -p 53
pktmon start --capture --comp nics --pkt-size 0 --file-name C:\Evidence\dc-locator.etl
Resolve-DnsName '_ldap._tcp.dc._msdcs.sbd.test' -Type SRV -Server 10.0.10.10
pktmon stop
pktmon etl2txt C:\Evidence\dc-locator.etl --out C:\Evidence\dc-locator.txt
pktmon etl2pcap C:\Evidence\dc-locator.etl --out C:\Evidence\dc-locator.pcapng
pktmon filter remove
```

Expect a query from 10.0.10.11 to 10.0.10.10 on UDP port 53 and an answer naming SBD-DC01 and SBD-DC02 on port 389. Read the text file, or open the .pcapng file in a packet analyzer such as Wireshark. Keep the capture with your evidence; it contains only the traffic the filter allowed.

SEE IT INSIDE THE NETWORK

Compare your capture with the packet card for Sign in, step 2.

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05 / EVIDENCE

Keep a useful build record

Use My evidence in the interactive guide to save chapter notes, confirm your checks, and answer each question. Export your record before moving to another browser or device.

Your evidence summary

Record the final domain and controllers, actual role holders, DNS resolver order, dated test output, and unresolved issues. Reference files stored in your own environment; do not record credentials.

RMF CONNECTION Implementation records what was configured. Assessment examines evidence. Monitoring revisits conclusions as the system changes.

Completion checklist

☐ I recorded actual settings and the reasons for them.

☐ I checked both DNS services and replication results.

☐ I kept dated evidence and described failures or limits.

☐ I listed recovery, hardening, and client checks still to do.

What did you observe

Evidence file names, collection time, key results, and the next action

GO FURTHER Explain why replication cannot recover every unwanted change. Identify a change that would trigger a new DNS or replication check.

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Continue the learning path

The workbook and interactive companion use the same five chapters. Your final recording will supply the original screenshots and video timestamps. Read the current sources before adapting the sample to another environment.

Technical guidance

Install Active Directory Domain Services

https://learn.microsoft.com/en-us/windows-server/identity/ad-ds/deploy/install-active-directory-domain-services--level-100-

DNS client configuration

https://learn.microsoft.com/en-us/troubleshoot/windows-server/networking/best-practices-for-dns-client-settings

DCDiag verification

https://learn.microsoft.com/en-us/windows-server/administration/windows-commands/dcdiag

Resolve-DnsName

https://learn.microsoft.com/en-us/powershell/module/dnsclient/resolve-dnsname?view=windowsserver2025-ps

Packet Monitor (pktmon)

https://learn.microsoft.com/en-us/windows-server/administration/windows-commands/pktmon

Service overview and network port requirements

https://learn.microsoft.com/en-us/troubleshoot/windows-server/networking/service-overview-and-network-port-requirements

Risk and control guidance

Risk Management Framework SP 800-37 Rev. 2

https://csrc.nist.gov/pubs/sp/800/37/r2/final

Security controls SP 800-53 Rev. 5

https://csrc.nist.gov/pubs/sp/800/53/r5/upd1/final

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